

Fabrication of high-refractive-index glass micron-sized solid immersion lenses by a surface-tension mold technique

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Abstract

A surface-tension mold technique that makes use of the wetting properties between glass melt and a substrate is used to fabricate micrometer-sized solid immersion lens from a high-refractive-index glass. Solid immersion lens can be used to overcome the optical diffraction limit. In this study, glass particles made from two different glasses ($n = 1.718$ and 1.786) were melted on glassy-carbon substrates having optical-grade surfaces and cooled to room temperature. The optimum conditions for producing a super-spherical shape (i.e., a truncated sphere) while avoiding the crystallization of glass were obtained. The produced glass particles exhibited high transparency and super-spherical shape with smooth surfaces. The variations in the lens thickness and sphericity are submicron or less. Optical lens systems specially designed for these super-spherical lenses are expected to be capable of achieving resolutions that are 2.0 times higher than those attainable with conventional optical systems.

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1. Introduction

Near-field optics has been receiving considerable attention recently; the main reason for this interest is that near-field optics can be used to achieve high spatial resolutions that exceed the optical diffraction limit. Several near-field techniques have been developed to reduce the optical spot size; solid immersion lenses (SILs) represent one such technique that holds out considerable promise and they have been investigated by many researchers [1–5]. Solid immersion lenses are generally hemispherical or super-spherical (i.e., a truncated sphere, the so-called Weierstrass sphere) in shape. The focal point of both kinds of SILs is

located on their bottom planar surface. When light reaches the base of the lens, an evanescent wave emerges from the bottom planar surface. The wavelength of the evanescent wave is reduced by a factor equal to the refractive index of the SIL, n , compared to that of the incident radiation. The spatial resolution of a hemispherical SIL (h -SIL) is improved by a factor of n , while that of a super-spherical SIL (super-SIL) is improved by a factor of n^2 due to an increase in its numerical aperture (NA) that occurs as a result of refraction at its spherical surface.

The shape of a near-field optical probe has to be strictly controlled in order to realize its theoretical performance. The thickness of a super-SIL, h , must be $a(1 + 1/n)$, where a is the radius of the spherical portion of the lens. Yoshita et al. [5] found expressions for the allowance for the aspheric error b and the thickness error d in SILs. In the case of super-SILs, b and d must lie within a few hundred nanome-

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ters of their theoretical values in order to realize the full potential of the lens. This requirement renders mass-fabrication of such high-precision super-spherical shapes impossible, since conventional techniques such as mechanical polishing are not capable of achieving such high precision.

Recently, we developed the surface-tension mold (StM) technique. This technique enables mass-fabrication of micrometer-sized super-SILs (μ -SILs) by making use of the wetting properties between the glass melt and a substrate [6]. Fine glass particles are deposited on a substrate and are then heat-treated at 800 °C in an H_2/N_2 atmosphere so that the particles become droplets. For a droplet having a size of the order of a micrometer, its shape on the substrate is determined by the balance between the three interfacial tensions since its weight is negligible. Thus, SILs fabricated by this technique are expected to have the same shapes. We prepared μ -SILs from soda-lime silicate glass ($n_d = 1.515$) and demonstrated that they can be used as evanescent-field optics that are capable of achieving horizontal resolutions that are 1.8 times higher than those of conventional lenses.

It is highly desirable to be able to produce high-refractive-index glass μ -SILs using the StM technique, since a SIL having a higher refractive index will have a higher spatial resolution. However, high-refractive-index glasses, especially Pb-free ones, have a strong tendency to crystallize when they are maintained at a high temperature. In order to prepare high-refractive-index glass μ -SILs, it is essential to determine suitable preparation conditions for the StM technique of such glasses.

In this study, we used the StM technique to fabricate super-spherical lenses from high-refractive-index glasses. We determined the conditions for fabricating super-spherical lenses without the glass crystallizing. We discuss the relationship between the obtained glass shape and the optical conditions for a SIL. Finally, a new design for a SIL system is proposed for utilizing the obtained shape of the super-spherical glass SIL.

2. Experimental procedures

Table 1 shows the compositions of the glasses used in this study (LZBS-1 and LZBS-2). They were developed as commercial optical glass formulations [7,8]. These glass compositions were selected because of their high-refractive-indices and high thermal stabilities, including resistance to crystallization. The glasses were prepared using a conventional melting and quenching method. The refractive indices of LZBS-1 and LZBS-2 glass at a wavelength of $\lambda = 663$ nm, n_{He-Ne} , were measured using an Abbe's refractometer and were found to be 1.718 and 1.786, respectively. Technical-grade reagents SiO_2 , H_3BO_3 , MgO , $CaCO_3$, $SrCO_3$, Li_2CO_3 , Na_2CO_3 , K_2CO_3 , ZnO , La_2O_3 , Gd_2O_3 , ZrO_2 and Ta_2O_5 were mixed and melted in a platinum crucible at 1050 °C for LZBS-1 glass and 1400 °C for LZBS-2 glass. The melting time was 1 h. In order to ensure that the glass was homogeneous, it was

Table 1
Nominal compositions of glasses (mol%) [7,8]

Glass	LZBS-1	LZBS-2
SiO_2	10.8	8.5
$B_2O_3^*$	35.2	30.9
MgO		3.6
CaO^*	9.4	
SrO^*		1.2
Li_2O^*	0.4	5.2
Na_2O^*	1.5	1.5
K_2O^*		0.8
ZnO	29.4	30.8
La_2O_3	13.3	7.8
Gd_2O_3		3.85
ZrO_2		1.2
Ta_2O_5		4.8

* These components were introduced as H_3BO_3 , $CaCO_3$, Li_2CO_3 , Na_2CO_3 and K_2CO_3 , respectively.

crushed and re-melted for another 3 h. The glass melt was poured onto a graphite plate, which was immediately transferred to an annealing furnace preheated at 500 °C. The glass sample was maintained at the annealing temperature for 1 h and then cooled to room temperature at a cooling rate of 1 °C/min. The obtained glass was crushed using an alumina mortar and classified using a sieve to obtain glass particles having grain sizes between 20 and 45 μ m.

Differential thermal analysis (DTA) was performed on the glass particles in air at a heating rate of 10 °C/min in order to estimate the glass transition temperature, T_g and T_b , which is located in the temperature range at which the glass particles begin to deform. The temperature accuracy was ± 5 °C.

The classified glass particles were dispersed on a finely polished glassy-carbon (g-c) substrate using CCl_4 . The particles were heated to 620–700 °C in an N_2 or H_2/N_2 atmosphere, and then cooled to room temperature. The heating and cooling rates were 10 °C/min. The obtained glass particles were observed using a scanning electron microscope (SEM) and an optical microscope (OM) to determine their shapes and transparencies.

3. Results

It was necessary to determine the characteristic temperatures of the glass samples in order to estimate the temperature range of the StM treatment for forming glass droplets on the substrate without causing the glass to crystallize. Fig. 1 shows the DTA curve for the LZBS-1 glass particles. The shallow endothermic change is attributed to T_g , and the subsequent one is determined as T_b . T_g and T_b were found to be 520 and 630 °C for LZBS-1 and 530 and 660 °C for LZBS-2, respectively. The exothermic peak of crystallization was not detected in either glass. Thus, we roughly estimated that the StM treatment can be performed on these glass particles at temperatures above T_b °C at a heating rate of 10 °C/min or greater.

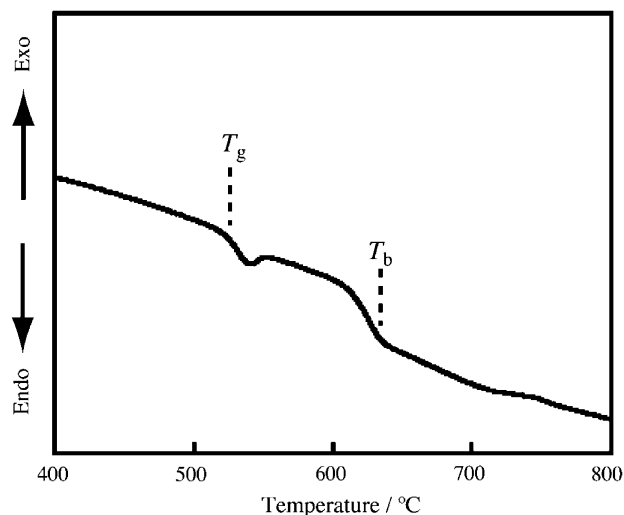


Fig. 1. DTA curve at a heating rate of 10 °C/min for LZBS-1 glass.

The results of the OM observation of the LZBS-1 glass particles heat-treated in an H_2/N_2 atmosphere are summarized in Fig. 2. The shapes of the glass droplets were unchanged in samples heat-treated below T_b ; however, glass particles heat-treated at temperatures above T_b were deformed. Samples heat-treated for long times exhibited surface crystallization. The dotted line in Fig. 2 indicates the boundary for crystallization (time–temperature–transformation (TTT) curves). Samples that exhibited partial crystallization were obtained above this line, but samples having a smooth plain surface were observed below this line. Thus, the shaded area in Fig. 2 corresponds to the applicable region of the StM treatment for preparing super-spherical glasses with high transparency.

Fig. 3 shows a typical OM photograph and SEM image of the sample heat-treated in N_2 . The sample in Fig. 3(a) has several ring patterns, although its surface is very smooth as shown in Fig. 3(b). These rings are due to holes formed in the base of the glass and observed through its

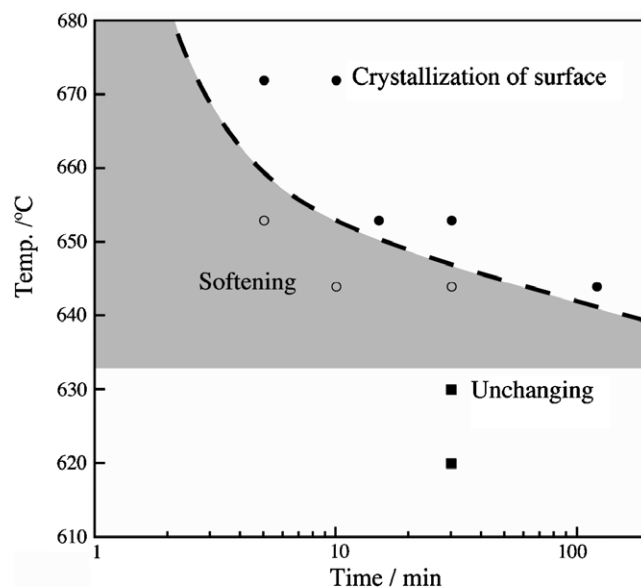


Fig. 2. Time–temperature–transformation diagram of LZBS-1 glass determined by OM observation. Softening with surface crystallization (filled circles), softening without crystallization (open circles) and no change in shape (filled squares) for heat-treatment under a H_2/N_2 atmosphere. The dotted line and shaded area are visual guides.

spherical surface. Such rings were observed in all samples heat-treated above T_b in an N_2 atmosphere for >5 min. The contact angles of particles having these holes were $<90^\circ$. The same phenomenon was also observed for the LZBS-2 glass.

Fig. 4(a) and (b) show SEM images of samples produced from LZBS-1 and LZBS-2, respectively. Both samples had defect-free surfaces, and the transparencies of both samples were confirmed by OM observations. The size of the obtained particles ranged from 20 to 50 μm . Their shapes commonly consisted of a curved surface and a plane one. The SEM observations ensured that the smooth curved surfaces had ellipticities of $<2\%$, which is the measuring

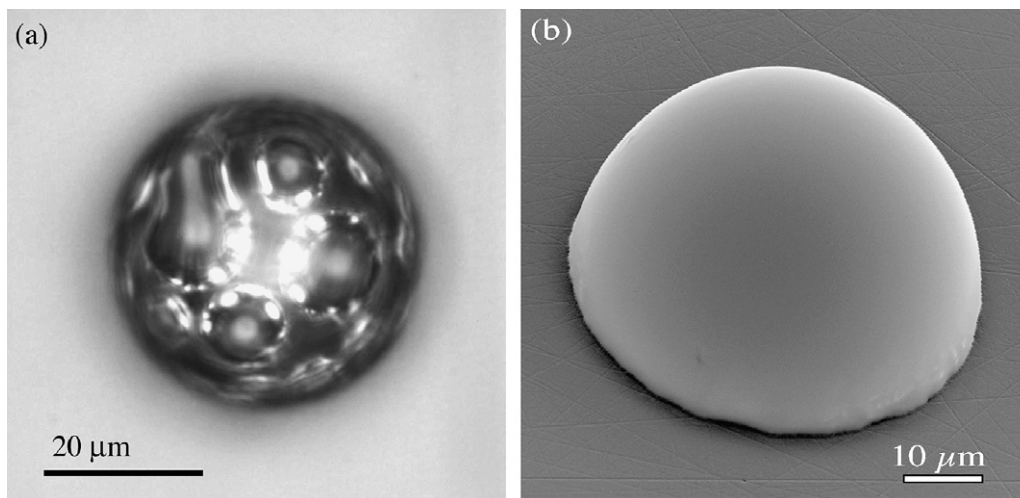


Fig. 3. (a) OM photograph and (b) SEM image of the sample made from LZBS-1. It was produced by heat-treatment under an N_2 atmosphere.

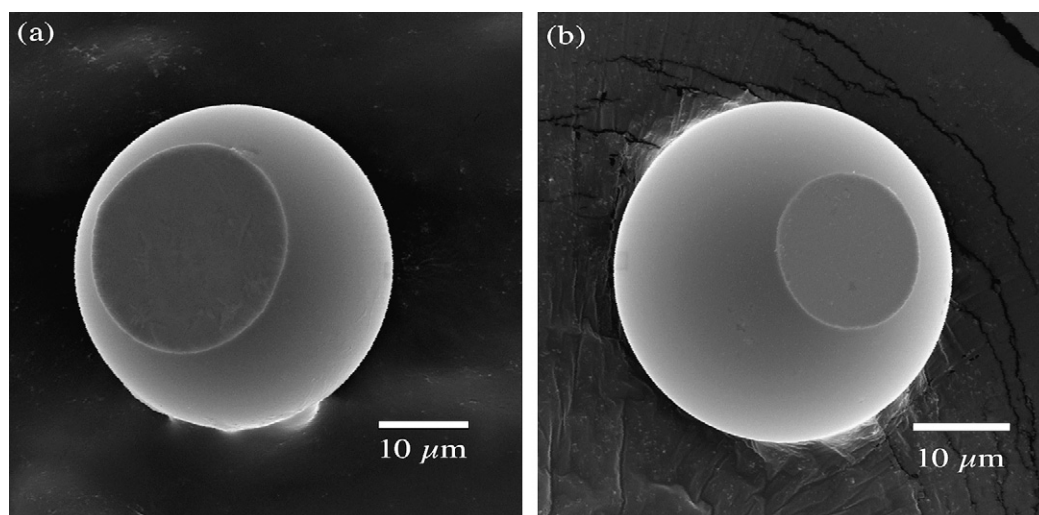


Fig. 4. SEM images of the samples produced from (a) LZBS-1 and (b) LZBS-2.

limit of the SEM observation due to image distortion. The plane surface was smooth, since it was formed by the glass melt making contact with the g-c substrate, which had an optical-grade surface. Heat treatment in H_2/N_2 did not result in the ring patterns that were observed in samples heat-treated in N_2 .

In the StM technique, high temperatures and long treatment times are preferable for attaining a high yield of well-controlled super-spherical droplets, because complete deformation of a super-spherical droplet requires considerable time. However, extreme conditions such as a high temperature or long treatment time result in crystallization. The heat-treatment programs were optimized for glass LZBS-1 and LZBS-2 using TTT diagrams such as the one shown in Fig. 2. The program pattern is shown in Fig. 5; the glass particles on the g-c substrate are heated to $T_2 = T_1 + 20^\circ\text{C}$ and immediately cooled to T_1 and then maintained at this temperature for 30 min, where the heating and cooling rates are $10^\circ\text{C}/\text{min}$. T_1 for LZBS-1 and LZBS-2 are 644 and 673°C , respectively.

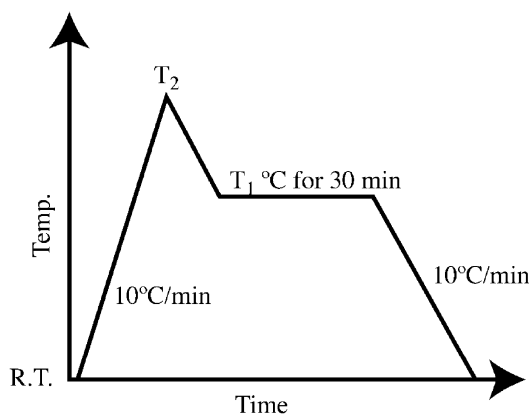


Fig. 5. Temperature program for LZBS-1 and LZBS-2. $T_2 = T_1 + 20^\circ\text{C}$. T_1 of LZBS-1 and LZBS-2 are 644 and 673°C , respectively.

The average contact angles θ of the super-spherical glass particles were estimated from the SEM images by using the radius of curvature of the spherical portion and the size of the bottom circular surface; they were found to be 137° and 152° for LZBS-1 and LZBS-2, respectively. The standard deviations of θ were $\pm 2^\circ$ of LZBS-1 and $\pm 1^\circ$ of LZBS-2, resulting in a scatter in the lens thickness Δh of 0.7 and $0.4\ \mu\text{m}$, respectively.

In order to realize its optical function, a super-SIL is required to have a refractive index that satisfies the condition,

$$n = -1/\cos \theta. \quad (1)$$

The obtained glass particles did not fully satisfy this condition. Based on Eq. (1) and the measured contact angles, the obtained super-spherical glasses of LZBS-1 and LZBS-2 require relative refractive indices of 1.37 and 1.13 , respectively (c.f., their measured refractive indices of 1.718 and 1.786 , respectively).

4. Discussion

The glass particles heat-treated in a N_2 atmosphere had holes at their bases (Fig. 3). These holes must be caused by the production of bubbles between the glass and the substrate at high temperatures. The produced bubbles produce holes in the contact line that delimits the wetted portion of the surface, making its contact angle comparatively low ($<90^\circ$). The gas that produces these bubbles has not been unambiguously identified, however, it is conjectured that the constituent of the gas is O_2 , N_2 or CO_x . Out of these gases, the holes are most likely to be generated by CO_x gas produced by the oxidation of carbon, since CO_2 has a lower solubility in borate glass melts than the other gases [9]. The reducing conditions in the H_2/N_2 atmosphere suppresses the oxidation of carbon and decreases the amount of CO_2 gas generated, so that hole defects are not observed.

Fig. 5 shows the temperature program that is applied to the glasses to produce a large number of super-spherical glass particles such as shown in Fig. 4. This program contains two stages to complete the formation of glass droplets on the substrate. The first stage is the formation of an initial droplet, the contact angle θ of which differs from the equilibrium contact angle. In the second stage, the line of contact moves along the substrate to approach the Young's contact angle determined by the wettability of the glass droplet with the g-c substrate. In the first stage, the major driving force that causes the viscous flow is the surface-tension of the liquid. The time taken for the initial droplets with smooth surfaces to deform is almost inversely proportional to the viscosity of the melt η , which increases exponentially with increasing temperature. Thus, time required to form the initial droplets can be accelerated remarkably by a slight increase in the heating temperature. On the other hand, the migration rate of the contact line does not depend only on η , but also on the difference between the apparent contact angle and the equilibrium one defined by Young's equation [10,11]. When the contact angle of the droplet is given by θ_a (termed the dynamic apparent contact angle), the force F at the contact line is given by

$$F = \gamma_{SL} - \gamma_S + \gamma_L \cos \theta_a, \quad (2)$$

where γ_{ij} is the interfacial tension between the media i and j , and the subscripts S and L denote the solid and liquid, respectively. If θ_a approaches the equilibrium value, F decreases to zero, resulting in a reduction in the deformation rate of the droplet. To avoid crystallization on the glass surface before the formation of a super-spherical shape, a lower temperature is necessary for the second step. According to the temperature program shown in Fig. 5, the initial droplet is formed quickly at T_2 , and then its contact line moves to attain its equilibrium shape at T_1 before crystallization.

We have succeeded in forming super-spherical glass lenses with a high-refractive-index and no crystallization, and their processing errors from the super-spherical shape were at least of submicron order. These values were considerably larger than the processing error required for near-field optical components such as SILs, but they were also lower than the errors produced in conventional mass-fabrication process. The reason for these errors in θ is due to the irregular shape of the particle prior to heat treatment. This is the origin of the contact angle hysteresis that causes the dispersion in the values of θ of the obtained glass particles.

In order to utilize the glass lenses that we produced as super-SILs for super-resolution optics, we have to design an optical system that satisfies Eq. (1). Eq. (1) states that the refractive index of the super-spherical glasses having θ of 137° and 152° should be 1.37 and 1.13, respectively. If the produced glasses are mounted in a transparent medium that has an appropriate refractive index, they will satisfy the condition for super-SILs. Fig. 6 shows one of the designs for SIL modules using a μ -SIL made from LZBS-2 glass; the LZBS-2 glass particle is mounted in a medium

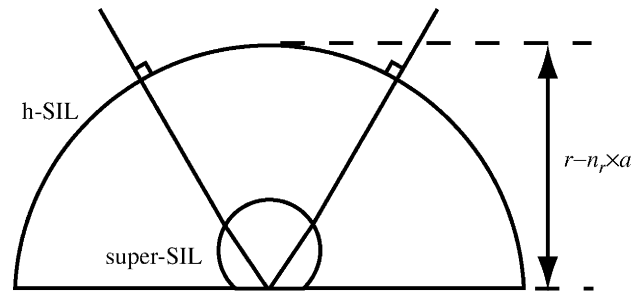


Fig. 6. Configuration of μ -SIL covered with h-SIL.

having a refractive index of 1.58, which results in a relative refractive index of 1.37. The thickness of the h-SIL is $r - n_r \times a$, where n_r is $n_{\mu\text{-SIL}}/n_{\text{h-SIL}}$, r and a are the radii of curvature of the h-SIL and the μ -SIL, respectively. The thickness allowance of the h-SIL is greater than the order of a micrometer, which is much larger than that of the super-SIL [12]. Thus, mass-fabrication of above-mentioned h-SIL component in the SIL module is straightforward. One candidate for materials for the h-SIL component is a plastic resin. This ensures that the NA by refraction at the surface of μ -SIL is $n_{\mu\text{-SIL}}/n_{\text{h-SIL}}$. The focal point is at the base of the μ -SIL, where the wavelength of the evanescent wave is $\lambda/n_{\mu\text{-SIL}}$. Thus, the spatial resolution is expected to be improved by a factor of $n_{\mu\text{-SIL}}^2/n_{\text{h-SIL}} = 2.0$.

5. Conclusion

A micrometer-size super-spherical lens of high-refractive-index glasses was prepared by the StM technique. The heating program was optimized for producing a super-spherical lens without glass crystallization. A new optical system for the solid immersion lens (SIL) designed, which allowed these lenses to achieve super-resolution. The fabricated lenses had an accuracy of at least submicron order, which is greater than the precision of conventional mass-fabrication process. These results demonstrate that the StM technique has the potential to realize mass-fabrication of SIL with high-refractive-index.

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